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Dielectric walls for complementary field effect transistors

Abstract

In an embodiment, a device includes: a dielectric wall; nanostructures abutting the dielectric wall; a lower source/drain region adjoining a lower subset of the nanostructures; an upper source/drain region adjoining an upper subset of the nanostructures, the upper source/drain region oppositely doped from the lower source/drain region; and a shared source/drain contact contacting the upper source/drain region and the lower source/drain region, the shared source/drain contact extending into the dielectric wall.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application claims the benefit of U.S. Provisional Application No. 63/421,320, filed on Nov. 1, 2022 and U.S. Provisional Application No. 63/374,024, filed on Aug. 31, 2022, which applications are hereby incorporated herein by reference.

BACKGROUND
(1) Semiconductor devices are used in a variety of electronic applications, such as, for example,

personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductor layers of material over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements thereon.
(2) The semiconductor industry continues to improve the integration density of various electronic components (e.g., transistors, diodes, resistors, capacitors, etc.) by continual reductions in minimum feature size, which allow more components to be integrated into a given area. However, as the minimum features sizes are reduced, additional problems arise that should be addressed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
(2) FIG. **1** illustrates an example of a nanostructure field-effect transistor (nanostructure-FET) in a three-dimensional view, in accordance with some embodiments.
(3) FIGS. **2-21**D are views of intermediate stages in the manufacturing of complementary field-effect transistors (complementary-FETs), in accordance with some embodiments.
(4) FIGS. **22-23** are views of complementary-FETs, in accordance with some embodiments.
(5) FIGS. **24**A-**24**C are views of complementary-FETs, in accordance with some embodiments.
(6) FIGS. **25**A-**25**C are views of complementary-FETs, in accordance with some embodiments.
(7) FIGS. **26**A-**26**C are views of complementary-FETs, in accordance with some embodiments.
(8) FIGS. **27-28** are views of complementary-FETs, in accordance with some embodiments.
(9) FIGS. **29**A-**30**C are views of additional intermediate stages in the manufacturing of complementary-FETs, in accordance with some embodiments.
(10) FIGS. **31**A-**31**D are views of complementary-FETs, in accordance with some embodiments.
(11) FIGS. **32-34** are views of intermediate stages in the manufacturing of complementary-FETs, in accordance with some embodiments.
DETAILED DESCRIPTION
(12) The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
(13) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted

accordingly.
(14) According to various embodiments, complementary-FETs include dielectric walls between adjacent nanostructures. The gate structures of the complementary-FETs are formed above the dielectric walls, and source/drain contacts for the complementary-FETs are formed partially in the dielectric walls. Because the gate structures are above the dielectric walls, a majority of a source/drain contact extends along a dielectric wall instead of along a gate structure. The parasitic capacitance between the source/drain contacts and the gate structures may thus be reduced.
(15) FIG. **1** illustrates an example of nanostructure-FETs (e.g., nanowire FETs, nanosheet FETs, multi bridge channel (MBC) FETs, nanoribbon FETs, gate-all-around (GAA) FETs, or the like), in accordance with some embodiments. FIG. **1** is a three-dimensional view, where some features of the nano structure-FETs are omitted for illustration clarity.
(16) As subsequently described in greater detail complementary-FETs will be formed from multiple vertically stacked nanostructure-FETs. A complementary-FET includes a lower nanostructure-FET of a first device type (e.g., n-type/p-type) and an upper nanostructure-FET of a second device type (e.g., p-type/n-type) that is opposite the first device type. FIG. **1** shows an example of lower nanostructure-FETs for the complementary-FETs.
(17) The nanostructure-FETs include nanostructures **66** (e.g., nano sheets, nanowires, or the like) over fins **62** on a substrate **50** (e.g., a semiconductor substrate), with the nanostructures **66** being semiconductor features that act as channel regions for the nanostructure-FETs. Isolation regions **76**, such as shallow trench isolation (STI) regions, may be disposed between adjacent fins **62**, which may protrude above and from between neighboring isolation regions **76**. The nanostructures **66** are disposed above and between adjacent isolation regions **76**. Although the isolation regions **76** are described/illustrated as being separate from the substrate **50**, as used herein, the term “substrate” may refer to the semiconductor substrate alone or a combination of the semiconductor substrate and the isolation regions. Additionally, although a bottom portion of the fins **62** are illustrated as being single, continuous materials with the substrate **50**, the bottom portion of the fins **62** and/or the substrate **50** may comprise a single material or a plurality of materials.
(18) Gate dielectrics **132** are over top surfaces of the fins **62** and along top surfaces, sidewalls, and bottom surfaces of the nanostructures **66**. Gate electrodes **134** are over the gate dielectrics **132**. Source/drain regions **108** are disposed on the fins **62** at opposing sides of the gate dielectrics **132** and the gate electrodes **134**. Source/drain region(s) **108** may refer to a source or a drain, individually or collectively dependent upon the context. An inter-layer dielectric (ILD) **114** is formed over the source/drain regions **108**.
(19) FIG. **1** further illustrates reference cross-sections that are used in later figures. Cross-section A-A′ is along a longitudinal axis of a fin **62** and in a direction of, for example, a current flow between the source/drain regions **108** of the devices. Cross-section B-B′ is perpendicular to cross-section A-A′ and is along a longitudinal axis of a gate electrode **134**. Cross-section C-C′ is parallel to cross-section B-B′ and extends through source/drain regions **108** of the devices.
(20) FIGS. **2**-**21**C are views of intermediate stages in the manufacturing of complementary-FETs, in accordance with some embodiments. FIGS. **2**, **3**, **4**, **5**, **6**, **7**, and **8** are three-dimensional views showing a similar three-dimensional view as FIG. **1**. FIGS. **9**A, **10**A, **11**A, **12**A, **13**A, **14**A, **15**A, **16**A, **17**A, **18**A, **19**A, **20**A, and **21**A illustrate cross-sectional views along a similar cross-section as reference cross-section A-A′ in FIG. **1**. FIGS. **9**B, **10**B, **11**B, **12**B, **13**B, **14**B, **15**B, **16**B, **17**B, **18**B, **19**B, **20**B, and **21**B illustrate cross-sectional views along a similar cross-section as reference cross-section B-B′ in FIG. **1**. FIGS. **9**C, **10**C, **11**C, **12**C, **13**C, **14**C, **15**C, **16**C, **17**C, **18**C, **19**C, **20**C, and **21**C illustrate cross-sectional views along a similar cross-section as reference cross-section C-C′ in FIG. **1**.
(21) In FIG. **2**, a substrate **50** is provided. The substrate **50** may be a semiconductor substrate, such as a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. The substrate **50** may be a wafer, such

as a silicon wafer. Generally, an SOI substrate is a layer of a semiconductor material formed on an insulator layer. The insulator layer may be, for example, a buried oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, typically a silicon or glass substrate. Other substrates, such as a multi-layered or gradient substrate may also be used. In some embodiments, the semiconductor material of the substrate **50** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including silicon-germanium, gallium arsenide phosphide, aluminum indium arsenide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and/or gallium indium arsenide phosphide; or combinations thereof.

(22) A multi-layer stack **52** is formed over the substrate **50**. The multi-layer stack **52** includes alternating first semiconductor layers **54** and second semiconductor layers **56**. The first semiconductor layers **54** are formed of a first semiconductor material, and the second semiconductor layers **56** are formed of a second semiconductor material. The semiconductor materials may each be selected from the candidate semiconductor materials of the substrate **50**.

(23) As subsequently described in greater detail, the first semiconductor layers **54** will be removed and the second semiconductor layers **56** will be patterned to form channel regions for the complementary-FETs. The first semiconductor layers **54** are dummy layers that will be removed in subsequent processing to expose the top surfaces and the bottom surfaces of the second semiconductor layers **56**. The first semiconductor material of the first semiconductor layers **54** is a material that has a high etching selectivity from the etching of the second semiconductor layers **56**, such as silicon-germanium. The second semiconductor material of the second semiconductor layers **56** is a material suitable for both n-type and p-type devices, such as silicon.

(24) The multi-layer stack **52** is illustrated as including six of the first semiconductor layers **54** and six of the second semiconductor layers **56**. It should be appreciated that the multi-layer stack **52** may include any number of the first semiconductor layers **54** and the second semiconductor layers **56**. Each of the layers of the multi-layer stack **52** may be grown by a process such as vapor phase epitaxy (VPE) or molecular beam epitaxy (MBE), deposited by a process such as chemical vapor deposition (CVD) or atomic layer deposition (ALD), or the like.

(25) In some embodiments, some layers of the multi-layer stack **52** are formed to be thicker than other layers of the multi-layer stack **52**. A middle semiconductor layer **54**B of the first semiconductor layers **54** may be thicker than others of the first semiconductor layers **54**A. A first subset of the second semiconductor layers **56** (e.g., lower semiconductor layers **56**L) are below the middle semiconductor layer **54**B. A second subset of the second semiconductor layers **56** (e.g., upper semiconductor layers **56**U) are above the middle semiconductor layer **54**B. The lower semiconductor layers **56**L may be formed of the same semiconductor material as the upper semiconductor layers **56**U, or may be formed of a different semiconductor material than the upper semiconductor layers **56**U.

(26) In FIG. **3**, fins **62** are formed in the substrate **50** and nanostructures **64**, **66** are formed in the multi-layer stack **52**. In some embodiments, the nanostructures **64**, **66** and the fins **62** may be formed in the multi-layer stack **52** and the substrate **50**, respectively, by etching trenches **60** in the multi-layer stack **52** and the substrate **50**. The etching may be any acceptable etch process, such as a reactive ion etch (RIE), neutral beam etch (NBE), the like, or a combination thereof. The etching may be anisotropic. Forming the nanostructures **64**, **66** by etching the multi-layer stack **52** may further define first nanostructures **64** from the first semiconductor layers **54** and define second nanostructures **66** from the second semiconductor layers **56**.

(27) The fins **62** and the nanostructures **64**, **66** may be patterned by any suitable method. For example, the fins **62** and the nanostructures **64**, **66** may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned

processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins **62** and the nanostructures **64**, **66**. In some embodiments, a mask (or other layer) may remain on the nanostructures **64**, **66**.

(28) Although each of the fins **62** and the nanostructures **64**, **66** are illustrated as having a constant width throughout, in other embodiments, the fins **62** and/or the nanostructures **64**, **66** may have tapered sidewalls such that a width of each of the fins **62** and/or the nanostructures **64**, **66** continuously increases in a direction towards the substrate **50**. In such embodiments, each of the nanostructures **64**, **66** may have a different width and be trapezoidal in shape.

(29) The thicknesses of the nanostructures **64**, **66** correspond to the thicknesses of the semiconductor layers **54**, **56**, respectively. As a result, the nanostructures **64** include isolation structures **64**B that are thicker than others of the nanostructures **64**A. The isolation structures **64**B will define the boundary of the devices of the complementary-FETs. A first, lower subset of the nanostructures **66** (e.g., lower nanostructures **66**L) are below the isolation structures **64**B. A second, upper subset of the nanostructures **66** (e.g., upper nanostructures **66**U) are above the isolation structures **64**B. The lower nanostructures **66**L will act as channel regions for lower nanostructure-FETs of the complementary-FETs, and the upper nanostructures **66**U will act as channel regions for upper nanostructure-FETs of the complementary-FETs.

(30) As subsequently described in greater detail for FIGS. **4-7**, dielectric walls **72** will be formed between some of the adjacent nanostructures **66**. The dielectric walls **72** will be formed in desired trenches **60** by forming dielectric walls **72** in all of the trenches **60**, and then removing a subset of the dielectric walls **72** such that only the dielectric walls **72** in the desired trenches **60** remain. Additionally, STI regions **76** will be formed in some or all of the trenches **60**. The STI regions **76** are different from the dielectric walls **72**. Although one dielectric wall **72** is illustrated, it should be appreciated that multiple dielectric walls **72** may be formed.

(31) In FIG. **4**, dielectric walls **72** are formed over the substrate **50** and in the trenches **60** between adjacent fins **62** and nanostructures **64**, **66**. The dielectric walls **72** may be formed of a dielectric material such as silicon oxide, silicon nitride, silicon oxynitride, silicon carbonitride, silicon oxycarbonitride, silicon oxycarbide, aluminum oxide, hafnium oxide, zirconium oxide, silicon carbide, combinations thereof, or the like, which may be deposited by chemical vapor deposition (CVD) or atomic layer deposition (ALD), or the like. In this embodiment, the dielectric walls **72** are single-layered, e.g., formed of a single layer of a single dielectric material. In another embodiment (subsequently described for FIGS. **25**A-**25**C), the dielectric walls **72** are multi-layered, e.g., formed of multiple layers of different dielectric materials.

(32) As an example to form the dielectric walls **72**, one or more of the previously described dielectric material(s) may be deposited in the trenches **60** between adjacent fins **62** and nanostructures **64**, **66**. The dielectric material(s) may also be deposited over the fins **62** and the nanostructures **64**, **66** such that excess dielectric material(s) cover the nanostructures **64**, **66**. A removal process is then applied to the dielectric material(s) to remove excess dielectric material(s) over the nanostructures **64**, **66**. In some embodiments, a planarization process such as a chemical mechanical polish (CMP), an etch-back process, combinations thereof, or the like may be utilized. The planarization process exposes the nanostructures **64**, **66** such that top surfaces of the nanostructures **64**, **66** and the dielectric material(s) are level after the planarization process is complete. In embodiments in which a mask remains on the nanostructures **64**, **66**, the planarization process may expose the mask or remove the mask such that top surfaces of the mask or the nanostructures **64**, **66**, respectively, and the dielectric material(s) are level after the planarization process is complete. The dielectric material(s), after the removal process, have portions left in the

trenches **60** (thus forming the dielectric walls **72**).
(33) In FIG. **5**, a first subset of the dielectric walls **72** are removed to re-form some of the trenches **60**. A second subset of the dielectric walls **72** are not removed, and remain in respective trenches **60**. The desired dielectric walls **72** may be removed using acceptable photolithography and etching techniques. For example, a mask (not separately illustrated) such as a photoresist may be formed over the dielectric walls **72** and the nanostructures **64**, **66**. The photoresist is patterned to expose the dielectric walls **72** that will be removed. The photoresist can be formed by using a spin-on technique and can be patterned using acceptable photolithography techniques. Once the photoresist is patterned, the dielectric walls **72** exposed by the photoresist may be removed by any acceptable etch process, such as one that is selective to the material(s) of the dielectric walls **72** (e.g., selectively etches the material(s) of the dielectric walls **72** at a faster rate than the materials of the nanostructures **64**, **66**). After the etch, the photoresist is removed, such as by an acceptable ashing process.
(34) In this embodiment, both sides of each dielectric wall **72** abut the adjacent nanostructures **66** after the removal process, such that the nanostructures **66** contact the sidewalls of the dielectric wall **72**. As a result, each dielectric wall **72** completely fills a trench **60** such that it extends continuously between the adjacent nanostructures **66**. In another embodiment (subsequently described for FIG. **27**), each dielectric wall **72** partially fills a trench **60** such that only one side of each dielectric wall **72** abuts the adjacent nanostructures **66**.
(35) In FIG. **6**, an insulation material **74** is formed over the substrate **50** and between some of the adjacent fins **62** and nanostructures **64**, **66**. Specifically, the insulation material **74** is in the trenches **60** between the adjacent fins **62** and nanostructures **64**, **66** where the dielectric walls **72** were removed. The insulation material **74** may be an oxide, such as silicon oxide, a nitride, the like, or a combination thereof, and may be formed by high-density plasma CVD (HDP-CVD), flowable CVD (FCVD), the like, or a combination thereof. Other insulation materials formed by any acceptable process may be used. The insulation material **74** is different from the material of the dielectric walls **72**. In some embodiments, the insulation material **74** includes silicon oxide formed by an FCVD process. An annealing process may be performed once the insulation material **74** is formed. Although the insulation material **74** is illustrated as a single layer, some embodiments may utilize multiple layers. For example, in some embodiments a liner (not separately illustrated) may first be formed along a surface of the substrate **50**, the fins **62**, and the nanostructures **64**, **66**. Thereafter, a fill material, such as one of the previously described insulation materials may be formed over the liner.
(36) The insulation material **74** may be deposited over the fins **62** and the nanostructures **64**, **66** such that excess insulation material **74** covers the nanostructures **64**, **66**. A removal process is then applied to the insulation material **74** to remove excess insulation material **74** over the nanostructures **64**, **66**. In some embodiments, a planarization process such as a chemical mechanical polish (CMP), an etch-back process, combinations thereof, or the like may be utilized. The planarization process exposes the nanostructures **64**, **66** such that top surfaces of the nanostructures **64**, **66** and the insulation material **74** are level after the planarization process is complete. In embodiments in which a mask remains on the nanostructures **64**, **66**, the planarization process may expose the mask or remove the mask such that top surfaces of the mask or the nanostructures **64**, **66**, respectively, and the insulation material **74** are level after the planarization process is complete.
(37) In FIG. **7**, the insulation material **74** is recessed to form STI regions **76**. The STI regions **76** are adjacent the fins **62**. The insulation material **74** is recessed such that upper portions of fins **62** and/or the nanostructures **64**, **66** protrude from between neighboring STI regions **76**. The upper portions of the fins **62** and/or the nanostructures **64**, **66** are above the STI regions **76**. Additionally, the upper portions of the dielectric walls **72** are above the STI regions **76**. Further, the top surfaces of the STI regions **76** may have a flat surface as illustrated, a convex surface, a concave surface

(such as dishing), or a combination thereof. The top surfaces of the STI regions **76** may be formed flat, convex, and/or concave by an appropriate etch. The STI regions **76** may be recessed using an acceptable etching process, such as one that is selective to the material of the insulation material **74** (e.g., etches the material of the insulation material **74** at a faster rate than the materials of the fins **62** and the nanostructures **64**, **66**). For example, an oxide removal using, for example, dilute hydrofluoric (dHF) acid may be used.

(38) The previously described process is just one example of how the fins **62** and the nanostructures **64**, **66** may be formed. In some embodiments, the fins **62** and/or the nanostructures **64**, **66** may be formed using a mask and an epitaxial growth process. For example, a dielectric layer can be formed over a top surface of the substrate **50**, and trenches can be etched through the dielectric layer to expose the underlying substrate **50**. Epitaxial structures can be epitaxially grown in the trenches, and the dielectric layer can be recessed such that the epitaxial structures protrude from the dielectric layer to form the fins **62** and/or the nanostructures **64**, **66**. The epitaxial structures may comprise the previously described alternating semiconductor materials, such as the first semiconductor materials and the second semiconductor materials. In some embodiments where epitaxial structures are epitaxially grown, the epitaxially grown materials may be in situ doped during growth, which may obviate prior and/or subsequent implantations, although in situ and implantation doping may be used together.

(39) In this embodiment, the dielectric walls **72** are formed before the STI regions **76**. As a result, the dielectric walls **72** contact a top surface of the substrate **50**. Additionally, the dielectric walls **72** are between a first subset of adjacent fins **62** and nanostructures **64**, **66** and the STI regions **76** are between a second subset of adjacent fins **62** and nanostructures **64**, **66**. In another embodiment (subsequently described for FIGS. **24**A-**24**C), the dielectric walls **72** are formed after the STI regions **76**, such that the dielectric walls **72** are formed above the STI regions **76**.

(40) Further, appropriate wells (not separately illustrated) may be formed in the fins **62**, the nanostructures **64**, **66**, the dielectric walls **72**, and/or the STI regions **76**. For example, an n-type impurity implant and/or a p-type impurity implant may be performed. The n-type impurities may be phosphorus, arsenic, antimony, or the like implanted in the region to a concentration in a range from 10.sup.17 atoms/cm.sup.3 to 10.sup.19 atoms/cm.sup.3. The p-type impurities may be boron, boron fluoride, indium, or the like implanted in the region to a concentration in a range from 10.sup.17 atoms/cm.sup.3 to 10.sup.19 atoms/cm.sup.3. When wells are formed in the nanostructures **66**, the wells may have a conductivity type opposite from a conductivity type of source/drain regions that will be subsequently formed adjacent the nanostructures **66**. Additionally, the wells in the lower nanostructures **66**L may have a conductivity type opposite from a conductivity type of the wells in the upper nanostructures **66**U. In some embodiments, the lower nanostructures **66**L have p-type wells and the upper nanostructures **66**U have n-type wells. In some embodiments, the lower nanostructures **66**L have n-type wells and the upper nanostructures **66**U have p-type wells. After the implant(s), an anneal may be performed to repair implant damage and to activate the p-type and/or n-type impurities that were implanted. In some embodiments, the grown materials of epitaxial fins may be in situ doped during growth, which may obviate the implantations, although in situ and implantation doping may be used together.

(41) In some embodiments, wells are formed in the fins **62** but not the nanostructures **64**, **66**. The wells in the fins **62** may have a conductivity type opposite from a conductivity type of source/drain regions that will be subsequently formed adjacent the lower nanostructures **66**L. In some embodiments, the fins **62** have n-type wells, p-type source/drain regions are subsequently formed adjacent the lower nanostructures **66**L, and n-type source/drain regions are subsequently formed adjacent the upper nanostructures **66**U. In some embodiments, the fins **62** have p-type wells, n-type source/drain regions are subsequently formed adjacent the lower nanostructures **66**L, and p-type source/drain regions are subsequently formed adjacent the upper nanostructures **66**U.

(42) In FIG. **8**, a dummy dielectric layer **82** is formed on the dielectric walls **72** and the fins **62**

and/or the nanostructures **64**, **66**. The dummy dielectric layer **82** may be formed of silicon oxide, silicon nitride, a combination thereof, or the like, which may be deposited or thermally grown according to acceptable techniques. A dummy gate layer **84** is formed over the dummy dielectric layer **82**, and a mask layer **86** is formed over the dummy gate layer **84**. The dummy gate layer **84** may be deposited over the dummy dielectric layer **82** and then planarized, such as by a CMP. The dummy gate layer **84** may be formed of a conductive or non-conductive material and may be selected from a group including amorphous silicon, polycrystalline-silicon (polysilicon), polycrystalline silicon-germanium (poly-SiGe), metallic nitrides, metallic silicides, metallic oxides, and metals. The material of the dummy gate layer **84** may be deposited by CVD, physical vapor deposition (PVD), sputter deposition, or other techniques for depositing the selected material. The dummy gate layer **84** may be formed of other materials that have a high etching selectivity from the etching of insulation materials, e.g., the STI regions **76** and/or the dummy dielectric layer **82**. The mask layer **86** may be deposited over the dummy gate layer **84**. The mask layer **86** may be formed of a dielectric material such as silicon nitride, silicon oxynitride, or the like. In the illustrated embodiment, the dummy dielectric layer **82** covers the STI regions **76** and the dielectric walls **72**, such that the dummy dielectric layer **82** extends between the dummy gate layer **84**, the STI regions **76**, and the dielectric walls **72**. In another embodiment, the dummy dielectric layer **82** covers only the fins **62** and/or the nanostructures **64**, **66**.

(43) In FIGS. **9**A-**9**C, the mask layer **86** is patterned using acceptable photolithography and etching techniques to form masks **96**. The pattern of the masks **96** then may be transferred to the dummy gate layer **84** and to the dummy dielectric layer **82** to form dummy gates **94** and dummy dielectrics **92**, respectively. The dummy gates **94** cover respective channel regions of the nanostructures **64**, **66**. The pattern of the masks **96** may be used to physically separate each of the dummy gates **94** from adjacent dummy gates **94**. The dummy gates **94** may also have a lengthwise direction substantially perpendicular to the lengthwise direction of respective fins **62**. The masks **96** can optionally be removed after patterning, such as by any acceptable etching technique.

(44) In FIGS. **10**A-**10**C, gate spacers **102** are formed over the nanostructures **64**, **66** and on exposed sidewalls of the masks **96** (if present), the dummy gates **94**, and the dummy dielectrics **92**. The gate spacers **102** may be formed by conformally forming one or more dielectric material(s) and subsequently etching the dielectric material(s). Acceptable dielectric materials may include silicon oxide, silicon nitride, silicon oxynitride, silicon oxycarbonitride, or the like, which may be formed by a deposition process such as chemical vapor deposition (CVD), atomic layer deposition (ALD), or the like. Other insulation materials formed by any acceptable process may be used. Any acceptable etch process, such as a dry etch, a wet etch, the like, or a combination thereof, may be performed to pattern the dielectric material(s). The etching may be anisotropic. The dielectric material(s), when etched, have portions left on the sidewalls of the dummy gates **94** (thus forming the gate spacers **102**).

(45) Further, implants for lightly doped source/drain (LDD) regions (not separately illustrated) may be performed. Appropriate type impurities may be implanted into the nanostructures **64**, **66**. The LDD regions may have a same conductivity type as a conductivity type of source/drain regions that will be subsequently formed adjacent the nanostructures **66**. Additionally, the LDD regions in the lower nanostructures **66**L may have a conductivity type opposite from a conductivity type of the LDD regions in the upper nanostructures **66**U. In some embodiments, the lower nanostructures **66**L have p-type LDD regions and the upper nanostructures **66**U have n-type LDD regions. In some embodiments, the lower nanostructures **66**L have n-type LDD regions and the upper nanostructures **66**U have p-type LDD regions. The n-type impurities may be the any of the n-type impurities previously discussed, and the p-type impurities may be the any of the p-type impurities previously discussed. The lightly doped source/drain regions may have a concentration of impurities in a range from 10.sup.17 atoms/cm.sup.3 to 10.sup.20 atoms/cm.sup.3. An anneal may be used to repair implant damage and to activate the implanted impurities.

(46) It is noted that the previous disclosure generally describes a process of forming spacers and LDD regions. Other processes and sequences may be used. For example, fewer or additional spacers may be utilized, different sequence of steps may be utilized, additional spacers may be formed and removed, and/or the like.
(47) Source/drain recesses **104** are patterned in the fins **62** and/or the nanostructures **64**, **66**. Source/drain regions will be subsequently formed in the source/drain recesses **104**. The source/drain recesses **104** may extend through the nanostructures **64**, **66** and into the substrate **50**. The source/drain recesses **104** extend through the lower nanostructures **66**L and through the upper nanostructures **66**U. In some embodiments, the fins **62** may be etched such that bottom surfaces of the source/drain recesses **104** are disposed below the top surfaces of the STI regions **76**. The source/drain recesses **104** may be formed by etching the fins **62** and/or the nanostructures **64**, **66** using anisotropic etching processes, such as RIE, NBE, or the like. The gate spacers **102** and the dummy gates **94** mask portions of the fins **62** and/or the nanostructures **64**, **66** during the etching processes used to form the source/drain recesses **104**. A single etch process or multiple etch processes may be used to etch each layer of the nanostructures **64**, **66** and/or the fins **62**. Timed etch processes may be used to stop the etching of the source/drain recesses **104** after the source/drain recesses **104** reach a desired depth.
(48) The dielectric walls **72** are exposed to the etchants utilized for patterning the source/drain recesses **104**. In this embodiment, no losses of the dielectric walls **72** occurs during the etching of the source/drain recesses **104**, such that the height of the dielectric walls **72** is not reduced. In another embodiment (subsequently described for FIGS. **22-23**), losses of the dielectric walls **72** occurs during the etching of the source/drain recesses **104**, such that the height of the dielectric walls **72** is reduced. The losses (if any) of the dielectric walls **72** depends on the etchants utilized for patterning the source/drain recesses **104**.
(49) In FIGS. **11**A-**11**C, inner spacers **106** are formed on the sidewalls of the remaining portions of the first nanostructures **64**, e.g., those sidewalls exposed by the source/drain recesses **104**. As subsequently described in greater detail, source/drain regions will be subsequently formed in the source/drain recesses **104**, and the first nanostructures **64** will be replaced with corresponding gate structures. The inner spacers **106** act as isolation features between the subsequently formed source/drain regions and the subsequently formed gate structures. Further, the inner spacers **106** may be used to prevent damage to the subsequently formed source/drain regions by subsequent etch processes, such as etch processes used to subsequently remove the first nanostructures **64**.
(50) As an example to form the inner spacers **106**, the source/drain recesses **104** can be laterally expanded. Specifically, portions of the sidewalls of the first nanostructures **64** exposed by the source/drain recesses **104** may be recessed to form sidewall recesses. Although sidewalls of the first nanostructures **64** are illustrated as being straight, the sidewalls may be concave or convex. The sidewalls may be recessed by any acceptable etch process, such as one that is selective to the material of the first nanostructures **64** (e.g., selectively etches the material of the first nanostructures **64** at a faster rate than the material of the second nanostructures **66**). The etching may be isotropic. For example, when the second nanostructures **66** are formed of silicon and the first nanostructures **64** are formed of silicon-germanium, the etch process may be a wet etch using tetramethylammonium hydroxide (TMAH), ammonium hydroxide (NH.sub.4OH), or the like. In another embodiment, the etch process may be a dry etch using a fluorine-based gas such as hydrogen fluoride (HF) gas. In some embodiments, the same etch process may be continually performed to both form the source/drain recesses **104** and recess the sidewalls of the first nanostructures **64**. The inner spacers **106** can then be formed by conformally forming an insulating material in the source/drain recesses **104**, and subsequently etching the insulating material. The insulating material may be silicon nitride or silicon oxynitride, although any suitable material, such as low-dielectric constant (low-k) materials having a k-value less than about 3.5, may be utilized. The insulating material may be formed by a deposition process, such as ALD, CVD, or the like.

The etching of the insulating material may be anisotropic. For example, the etch process may be a dry etch such as a RIE, a NBE, or the like.
(51) Although outer sidewalls of inner spacers **106** are illustrated as being flush with sidewalls of the second nanostructures **66**, the outer sidewalls of the inner spacers **106** may extend beyond or be recessed from sidewalls of the second nanostructures **66**. In other words, the inner spacers **106** may partially fill, completely fill, or overfill the sidewall recesses. Moreover, although the sidewalls of the inner spacers **106** are illustrated as being straight, the sidewalls of the inner spacers **106** may be concave or convex.
(52) Lower epitaxial source/drain regions **108** are formed in the lower portions of the source/drain recesses **104**. The lower epitaxial source/drain regions **108** only partially fill the source/drain recesses **104**, such that the lower epitaxial source/drain regions **108** are in contact with the lower nanostructures **66**L and are not in contact with the upper nanostructures **66**U. In some embodiments, the lower epitaxial source/drain regions **108** exert stress in the respective channel regions of the lower nanostructures **66**L, thereby improving performance. The lower epitaxial source/drain regions **108** are formed in the source/drain recesses **104** such that each dummy gate **94** is disposed between respective neighboring pairs of the lower epitaxial source/drain regions **108** in a top-down view. In some embodiments, the inner spacers **106** are used to separate the lower epitaxial source/drain regions **108** from the nanostructures **64** by an appropriate lateral distance so that the lower epitaxial source/drain regions **108** do not short out with subsequently formed gates of the resulting devices.
(53) The lower epitaxial source/drain regions **108** are epitaxially grown in the lower portions of the source/drain recesses **104**. The height of the lower epitaxial source/drain regions **108** is less than the distance between the fins **62** and the upper nanostructures **66**U. Timed epitaxial growth processes may be used to stop the growth of the lower epitaxial source/drain regions **108** after the lower epitaxial source/drain regions **108** reach a desired height. The lower epitaxial source/drain regions **108** may have any desired conductivity type, which is opposite the conductivity type of the channel regions of the lower nanostructures **66**L. In some embodiments, the lower epitaxial source/drain regions **108** may be in situ doped during growth.
(54) In some embodiments, the lower epitaxial source/drain regions **108** are n-type source/drain regions. For example, if the lower nanostructures **66**L are silicon, the lower epitaxial source/drain regions **108** may include materials exerting a tensile strain on the lower nanostructures **66**L, such as silicon, silicon carbide, phosphorous-doped silicon carbide, silicon phosphide, silicon arsenide, or the like. The lower epitaxial source/drain regions **108** may have surfaces raised from respective upper surfaces of the lower nanostructures **66**L and may have facets.
(55) In some embodiments, the lower epitaxial source/drain regions **108** are p-type source/drain regions. For example, if the lower nanostructures **66**L are silicon-germanium, the lower epitaxial source/drain regions **108** may include materials exerting a compressive strain on the lower nanostructures **66**L, such as silicon-germanium, boron-doped silicon-germanium, boron-doped silicon, germanium, germanium tin, or the like. The lower epitaxial source/drain regions **108** may have surfaces raised from respective upper surfaces of the lower nanostructures **66**L and may have facets.
(56) The lower epitaxial source/drain regions **108** may be implanted with dopants to form source/drain regions, similar to the process previously discussed for forming lightly-doped source/drain regions, followed by an anneal. The source/drain regions may have an impurity concentration in the range of 10.sup.19 atoms/cm.sup.3 and 10.sup.21 atoms/cm.sup.3. The n-type and/or p-type impurities for source/drain regions may be any of the impurities previously discussed. In some embodiments, the lower epitaxial source/drain regions **108** are in situ doped during growth.
(57) As a result of the epitaxy processes used to form the lower epitaxial source/drain regions **108**, upper surfaces of the lower epitaxial source/drain regions **108** have facets which expand laterally

outward beyond sidewalls of the lower nanostructures **66**L. In some embodiments, the dielectric walls **72** block the epitaxial growth, such that the lower epitaxial source/drain regions **108** at opposing sides of the dielectric walls **72** remain separated after the epitaxy process is completed. In some embodiments, fin spacers (not separately illustrated) are formed covering a portion of the sidewalls of the fins **62** that extend above the STI regions **76**, thereby blocking the epitaxial growth. In some embodiments, the lower epitaxial source/drain regions **108** extend to the surface of the STI regions **76**.

(58) The lower epitaxial source/drain regions **108** may comprise one or more semiconductor material layers. For example, the lower epitaxial source/drain regions **108** may comprise a liner layer, a main layer, and a finishing layer (or more generally, a first semiconductor material layer, a second semiconductor material layer, and a third semiconductor material layer). Each of the liner layer, the main layer, and the finishing layer may be formed of different semiconductor materials and may be doped to different dopant concentrations. In embodiments in which the lower epitaxial source/drain regions **108** include three semiconductor material layers, the liner layers may be grown in the source/drain recesses **104**, the main layers may be grown on the liner layers, and the finishing layers may be grown on the main layers. Any number of semiconductor material layers may be used for the lower epitaxial source/drain regions **108**.

(59) In FIGS. **12**A-**12**C, a first ILD **114** is formed over the lower epitaxial source/drain regions **108**, the gate spacers **102**, the masks **96** (if present) or the dummy gates **94**, the STI regions **76**, and the dielectric walls **72**. The first ILD **114** may be formed of a dielectric material, which may be deposited by any suitable method, such as CVD, plasma-enhanced CVD (PECVD), or FCVD. Dielectric materials may include phospho-silicate glass (PSG), boro-silicate glass (BSG), boron-doped phospho-silicate glass (BPSG), undoped silicate glass (USG), or the like. Other insulation materials formed by any acceptable process may be used.

(60) In some embodiments, a first contact etch stop layer (CESL) **112** is formed between the first ILD **114** and the lower epitaxial source/drain regions **108**, the gate spacers **102**, the masks **96** (if present) or the dummy gates **94**, the STI regions **76**, and the dielectric walls **72**. The first CESL **112** may be formed of a dielectric material having a high etching selectivity from the etching of the first ILD **114**, such as silicon nitride, silicon oxide, silicon oxynitride, or the like, which may be formed by any suitable deposition process, such as CVD, ALD, or the like.

(61) In FIG. **13**A-**13**C, the first ILD **114** and the first CESL **112** are recessed to re-form the upper portions of the source/drain recesses **104**. After the recessing, the first CESL **112** and the first ILD **114** only partially fill the source/drain recesses **104**, such that the sidewalls of the upper nanostructures **66**U are exposed. Upper portions of the dielectric walls **72** may also be exposed. The first ILD **114** and the first CESL **112** may be recessed using an acceptable etching process. The etching exposes the sidewalls of the upper nanostructures **66**U. In some embodiments, the first ILD **114** is etched using the first CESL **112** as an etch stop layer, and the first CESL **112** is then etched using the first ILD **114** as an etching mask.

(62) Upper epitaxial source/drain regions **118** are then formed in the upper portions of the source/drain recesses **104**. The upper epitaxial source/drain regions **118** only partially fill the source/drain recesses **104**, such that the upper epitaxial source/drain regions **118** are in contact with the upper nanostructures **66**U and are not in contact with the lower nanostructures **66**L. In some embodiments, the upper epitaxial source/drain regions **118** exert stress in the respective channel regions of the upper nanostructures **66**U, thereby improving performance. The upper epitaxial source/drain regions **118** are formed in the source/drain recesses **104** such that each dummy gate **94** is disposed between respective neighboring pairs of the upper epitaxial source/drain regions **118** in a top-down view. In some embodiments, the inner spacers **106** are used to separate the upper epitaxial source/drain regions **118** from the nanostructures **64** by an appropriate lateral distance so that the upper epitaxial source/drain regions **118** do not short out with subsequently formed gates of the resulting devices.

(63) The upper epitaxial source/drain regions **118** are epitaxially grown in the upper portions of the source/drain recesses **104**. The upper epitaxial source/drain regions **118** may have any desired conductivity type, which is opposite the conductivity type of the channel regions of the upper nanostructures **66**U. The conductivity type of the upper epitaxial source/drain regions **118** may be opposite the conductivity type of the lower epitaxial source/drain regions **108**, and the conductivity type of the channel regions of the upper nanostructures **66**U may be opposite the conductivity type of the channel regions of the lower nanostructures **66**L. Put another way, the upper epitaxial source/drain regions **118** may be oppositely doped from the lower epitaxial source/drain regions **108**, and the upper nanostructures **66**U may be oppositely doped from the lower nanostructures **66**L. In some embodiments, the upper epitaxial source/drain regions **118** may be in situ doped during growth.
(64) In some embodiments, the upper epitaxial source/drain regions **118** are n-type source/drain regions. For example, if the upper nanostructures **66**U are silicon, the upper epitaxial source/drain regions **118** may include materials exerting a tensile strain on the upper nanostructures **66**U, such as silicon, silicon carbide, phosphorous-doped silicon carbide, silicon phosphide, silicon arsenide, or the like. The upper epitaxial source/drain regions **118** may have surfaces raised from respective upper surfaces of the upper nanostructures **66**U and may have facets.
(65) In some embodiments, the upper epitaxial source/drain regions **118** are p-type source/drain regions. For example, if the upper nanostructures **66**U are silicon-germanium, the upper epitaxial source/drain regions **118** may include materials exerting a compressive strain on the upper nanostructures **66**U, such as silicon-germanium, boron-doped silicon-germanium, boron-doped silicon, germanium, germanium tin, or the like. The upper epitaxial source/drain regions **118** may have surfaces raised from respective upper surfaces of the upper nanostructures **66**U and may have facets.
(66) The upper epitaxial source/drain regions **118** may be implanted with dopants to form source/drain regions, similar to the process previously discussed for forming lightly-doped source/drain regions, followed by an anneal. The source/drain regions may have an impurity concentration in the range of 10.sup.19 atoms/cm.sup.3 and 10.sup.21 atoms/cm.sup.3. The n-type and/or p-type impurities for source/drain regions may be any of the impurities previously discussed. In some embodiments, the upper epitaxial source/drain regions **118** are in situ doped during growth.
(67) As a result of the epitaxy processes used to form the upper epitaxial source/drain regions **118**, upper surfaces of the upper epitaxial source/drain regions **118** have facets which expand laterally outward beyond sidewalls of the upper nanostructures **66**U. In some embodiments, the dielectric walls **72** block the epitaxial growth, such that the upper epitaxial source/drain regions **118** at opposing sides of the dielectric walls **72** remain separated after the epitaxy process is completed.
(68) The upper epitaxial source/drain regions **118** may comprise one or more semiconductor material layers. For example, the upper epitaxial source/drain regions **118** may comprise a liner layer, a main layer, and a finishing layer (or more generally, a first semiconductor material layer, a second semiconductor material layer, and a third semiconductor material layer). Each of the liner layer, the main layer, and the finishing layer may be formed of different semiconductor materials and may be doped to different dopant concentrations. In embodiments in which the upper epitaxial source/drain regions **118** include three semiconductor material layers, the liner layers may be grown in the source/drain recesses **104**, the main layers may be grown on the liner layers, and the finishing layers may be grown on the main layers. Any number of semiconductor material layers may be used for the upper epitaxial source/drain regions **118**.
(69) The source/drain recesses **104** may be completely filled by the combination of the upper epitaxial source/drain regions **118**, the first ILD **114**, the first CESL **112**, and the lower epitaxial source/drain regions **108**. The first ILD **114** and the first CESL **112** are between the upper epitaxial source/drain regions **118** and the lower epitaxial source/drain regions **108**. The lower epitaxial

source/drain regions **108** are for lower nanostructure-FETs of complementary-FETs, and the upper epitaxial source/drain regions **118** are for upper nanostructure-FETs of the complementary-FETs. The first ILD **114** and the first CESL **112** thus act as isolation regions to prevent shorting of the lower and upper nanostructure-FETs.
(70) In FIGS. **14**A-**14**C, a second ILD **124** is deposited over the upper epitaxial source/drain regions **118**, the first ILD **114**, the gate spacers **102**, and the masks **96** (if present) or the dummy gates **94**. The second ILD **124** may be formed of a dielectric material, which may be deposited by any suitable method, such as CVD, plasma-enhanced CVD (PECVD), or FCVD. Dielectric materials may include phospho-silicate glass (PSG), boro-silicate glass (BSG), boron-doped phospho-silicate glass (BPSG), undoped silicate glass (USG), or the like. Other insulation materials formed by any acceptable process may be used.
(71) In some embodiments, an second CESL **122** is formed between the second ILD **124** and the upper epitaxial source/drain regions **118**, the first ILD **114**, the gate spacers **102**, and the masks **96** (if present) or the dummy gates **94**. The second CESL **122** may be formed of a dielectric material having a high etching selectivity from the etching of the second ILD **124**, such as, silicon nitride, silicon oxide, silicon oxynitride, or the like, which may be formed by any suitable deposition process, such as CVD, ALD, or the like.
(72) In FIGS. **15**A-**15**C, a removal process is performed to level the top surfaces of the second ILD **124** with the top surfaces of the gate spacers **102** and the masks **96** (if present) or the dummy gates **94**. In some embodiments, a planarization process such as a chemical mechanical polish (CMP), an etch-back process, combinations thereof, or the like may be utilized. The planarization process may also remove the masks **96** on the dummy gates **94**, and portions of the gate spacers **102** along sidewalls of the masks **96**. After the planarization process, top surfaces of the second ILD **124**, the second CESL **122**, the gate spacers **102**, and the masks **96** (if present) or the dummy gates **94** are substantially coplanar (within process variations). Accordingly, the top surfaces of the masks **96** (if present) or the dummy gates **94** are exposed through the second ILD **124**. In the illustrated embodiment, the masks **96** are removed such that the top surfaces of the dummy gates **94** are exposed through the second ILD **124**.
(73) In FIGS. **16**A-**16**C, the dummy gates **94** are removed in one or more etching steps, so that recesses **126** are formed between the gate spacers **102**. Portions of the dummy dielectrics **92** in the recesses **126** are also removed. In some embodiments, the dummy gates **94** and the dummy dielectrics **92** are removed by an anisotropic dry etch process. For example, the etching process may include a dry etch process using reaction gas(es) that selectively etch the material of the dummy gates **94** at a faster rate than the materials of the second ILD **124** and the gate spacers **102**. Each recesses **126** exposes and/or overlies portions of nanostructures **64**, **66**, which act as the channel regions in the resulting devices. Portions of the nanostructures **64**, **66** which act as the channel regions are disposed between neighboring pairs of the lower epitaxial source/drain regions **108** and the upper epitaxial source/drain regions **118**. During the removal, the dummy dielectrics **92** may be used as etch stop layers when the dummy gates **94** are etched. The dummy dielectrics **92** may then be removed after the removal of the dummy gates **94**.
(74) The remaining portions of the first nanostructures **64** are then removed to form openings **128** in regions between the second nanostructures **66**. The remaining portions of the first nanostructures **64** can be removed by any acceptable etch process that selectively etches the material of the first nanostructures **64** at a faster rate than the material of the second nanostructures **66**. The etching may be isotropic. For example, when the first nanostructures **64** are formed of silicon-germanium and the second nanostructures **66** are formed of silicon, the etch process may be a wet etch using tetramethylammonium hydroxide (TMAH), ammonium hydroxide (NH.sub.4OH), or the like. In some embodiments, a trim process (not separately illustrated) is performed to decrease the thicknesses of the exposed portions of the second nanostructures **66** and expand the openings **128**.
(75) In this embodiment, no etching of the dielectric walls **72** occurs during the formation of the

openings **128**. As a result, the openings **128** do not extend into the dielectric walls **72**. In another embodiment (subsequently described for FIGS. **26**A-**26**C), some etching of the dielectric walls **72** occurs during the formation of the openings **128**.

(76) In FIGS. **17**A-**17**C, gate dielectrics **132** and gate electrodes **134** are formed for replacement gates. Each respective pair of a gate dielectric **132** and a gate electrode **134** may be collectively referred to as a "gate structure." Each gate structure extends along three sides (e.g., a top surface, a sidewall, and a bottom surface) of a channel region of a nanostructure **66**, such that the gate structure extends along sidewalls, a bottom surface, and a top surface of the nanostructure **66**. The gate structures also extend along top surfaces and sidewalls of the dielectric walls **72**. The gate structures may also extend along sidewalls and/or a top surface of a fin **62**.

(77) The gate dielectrics **132** include one or more gate dielectric layer(s) disposed on the sidewalls and/or the top surfaces of the fins **62**; on the top surfaces, the sidewalls, and the bottom surfaces of the channel regions of the nanostructures **66**; on the sidewalls of the inner spacers **106**; on the sidewalls of the gate spacers **102**; and on the sidewalls and/or the top surfaces of the dielectric walls **72**. The gate dielectrics **132** may be formed of an oxide such as silicon oxide or a metal oxide, a silicate such as a metal silicate, combinations thereof, multi-layers thereof, or the like. Additionally or alternatively, the gate dielectrics **132** may be formed of a high-k dielectric material (e.g., dielectric materials having a k-value greater than about 7.0), such as a metal oxide and/or silicate of hafnium, aluminum, zirconium, lanthanum, manganese, barium, titanium, lead, and combinations thereof. The dielectric material(s) of the gate dielectrics **132** may be formed by molecular-beam deposition (MBD), ALD, PECVD, or the like. Although single-layered gate dielectrics **132** are illustrated, the gate dielectrics **132** may include any number of interfacial layers and any number of main layers. For example, the gate dielectrics **132** may include an interfacial layer and an overlying high-k dielectric layer.

(78) The gate electrodes **134** include one or more gate electrode layer(s) disposed over the gate dielectrics **132**. The gate electrodes **134** may be formed of a metal-containing material such as tungsten, titanium, titanium nitride, tantalum, tantalum nitride, tantalum carbide, aluminum, ruthenium, cobalt, combinations thereof, multi-layers thereof, or the like. Although single-layered gate electrodes **134** are illustrated, the gate electrodes **134** may include any number of work function tuning layers, any number of barrier layers, any number of glue layers, and a fill material.

(79) As an example to form the gate structures, one or more gate dielectric layer(s) may be deposited in the recesses **126** and the openings **128**. The gate dielectric layer(s) may also be deposited on the top surfaces of the second ILD **124**, the second CESL **122**, the gate spacers **102**, and the dielectric walls **72**. Subsequently, one or more gate electrode layer(s) may be deposited on the gate dielectric layer(s), and in the remaining portions of the recesses **126** and the openings **128**. A removal process may then be performed to remove the excess portions of the gate dielectric layer(s) and the gate electrode layer(s), which excess portions are over the top surfaces of the second ILD **124**, the second CESL **122**, and the gate spacers **102**. In some embodiments, a planarization process such as a chemical mechanical polish (CMP), an etch-back process, combinations thereof, or the like may be utilized. The gate dielectric layer(s), after the removal process, have portions left in the recesses **126** and the openings **128** (thus forming the gate dielectrics **132**). The gate electrode layer(s), after the removal process, have portions left in the recesses **126** and the openings **128** (thus forming the gate electrodes **134**). When a planarization process is utilized, the top surfaces of the gate spacers **102**, the second CESL **122**, the second ILD **124**, the gate dielectrics **132**, and the gate electrodes **134** are substantially coplanar (within process variations).

(80) In this embodiment where no losses of the dielectric walls **72** occurs during the etching of the source/drain recesses **104** (see FIGS. **10**A-**10**C), the portions of the dielectric walls **72** beneath the gate structures (see FIG. **17**B) have a same height as the portions of the dielectric walls **72** adjacent the lower epitaxial source/drain regions **108** and the upper epitaxial source/drain regions **118** (see

FIG. **17**C). Additionally, in this embodiment, the top surfaces of the gate electrodes **134** are above the top surfaces of the dielectric walls **72**. In another embodiment (subsequently described for FIGS. **32-34**), the top surfaces of the gate electrodes **134** are coplanar with top surfaces of the dielectric walls **72**.

(81) In FIGS. **18**A-**18**C, a gate isolation structure **136** is formed to divide (or “cut”) a gate structure (including a gate dielectric **132** and a gate electrode **134**) into multiple gate segments. As an example to form the gate isolation structure **136**, an opening may be patterned in a gate structure. Any acceptable etching process, such as a dry etch, a wet etch, the like, or a combination thereof, may be performed to pattern the opening. The etching may be anisotropic. The opening exposes a top surface of a dielectric wall **72**. One or more dielectric material(s) are deposited in the opening. Acceptable dielectric materials include silicon nitride, silicon oxide, silicon oxynitride, or the like, which may be formed by a deposition process such as CVD, ALD, or the like. A removal process may be performed to remove the excess portions of the dielectric material(s), which excess portions are over the top surfaces of the gate electrode **134**, thereby forming the gate isolation structure **136**. A gate isolation structure **136** may isolate the gate structures of adjacent devices.

(82) Contact openings **142** are formed to expose the lower epitaxial source/drain regions **108** and the upper epitaxial source/drain regions **118**. The contact openings **142** expose top surfaces and sidewalls of the upper epitaxial source/drain regions **118**, sidewalls of the lower epitaxial source/drain regions **108**, and sidewalls of the dielectric walls **72**. The contact openings **142** may be formed using acceptable photolithography and etching techniques.

(83) In some embodiments, a multi-step etching process is utilized to form the contact openings **142**. A first etch may be performed to form upper portions of the contact openings **142** in the second ILD **124** and the second CESL **122**, thereby exposing top surfaces of the upper epitaxial source/drain regions **118** and the dielectric walls **72**. The first etch may be selective to the materials of the second ILD **124** and the second CESL **122** (e.g., selectively etches the materials of the second ILD **124** and the second CESL **122** at a faster rate than the material(s) of the dielectric walls **72**). The etching may be anisotropic. A second etch may then be performed to form the lower portions of the contact openings **142** in the dielectric walls **72**, thereby exposing the sidewalls of the lower epitaxial source/drain regions **108** and the sidewalls of the upper epitaxial source/drain regions **118**. The second etch may be selective to the material(s) of the dielectric walls **72** (e.g., selectively etches the material(s) of the dielectric walls **72** at a faster rate than the materials of the lower epitaxial source/drain regions **108** and the upper epitaxial source/drain regions **118**).

(84) In FIGS. **19**A-**19**C, shared source/drain contacts **146** are formed in the contact openings **142**. The shared source/drain contacts **146** are physically and electrically coupled to the lower epitaxial source/drain regions **108** and to the upper epitaxial source/drain regions **118**. Specifically, a shared source/drain contact **146** is shared with both an upper epitaxial source/drain region **118** and a lower epitaxial source/drain region **108** that is below the upper epitaxial source/drain region **118**.

(85) As an example to form the shared source/drain contacts **146**, a liner (not separately illustrated), such as a diffusion barrier layer, an adhesion layer, or the like, and a conductive material are formed in the contact openings **142**. The liner may include titanium, titanium nitride, tantalum, tantalum nitride, or the like. The conductive material may be cobalt, tungsten, copper, a copper alloy, silver, gold, aluminum, nickel, or the like. A removal process may be performed to remove excess material from the top surfaces of the gate spacers **102**, the second ILD **124**, the gate electrodes **134**, and the gate isolation structure **136**. The remaining liner and conductive material form the shared source/drain contacts **146** in the contact openings **142**. In some embodiments, a planarization process such as a chemical mechanical polish (CMP), an etch-back process, combinations thereof, or the like is utilized. After the planarization process, the top surfaces of the gate spacers **102**, the second ILD **124**, the gate electrodes **134**, the gate isolation structure **136**, and the shared source/drain contacts **146** are substantially coplanar (within process variations).

(86) Optionally, metal-semiconductor alloy regions **144** are formed at the interfaces between the

shared source/drain contacts **146** and the lower epitaxial source/drain regions **108** and/or to the upper epitaxial source/drain regions **118**. The metal-semiconductor alloy regions **144** can be silicide regions formed of a metal silicide (e.g., titanium silicide, cobalt silicide, nickel silicide, etc.), germanide regions formed of a metal germanide (e.g. titanium germanide, cobalt germanide, nickel germanide, etc.), silicon-germanide regions formed of both a metal silicide and a metal germanide, or the like. The metal-semiconductor alloy regions **144** can be formed before the shared source/drain contacts **146** by depositing a metal in the contact openings **142** and then performing a thermal anneal process. The metal can be any metal capable of reacting with the semiconductor materials (e.g., silicon, silicon-germanium, germanium, etc.) of the lower epitaxial source/drain regions **108** and/or to the upper epitaxial source/drain regions **118** to form a low-resistance metal-semiconductor alloy, such as nickel, cobalt, titanium, tantalum, platinum, tungsten, other noble metals, other refractory metals, rare earth metals or their alloys. The metal can be deposited by a deposition process such as ALD, CVD, PVD, or the like. After the thermal anneal process, a cleaning process, such as a wet clean, may be performed to remove any residual metal from the contact openings **142**, such as from surfaces of the metal-semiconductor alloy regions **144**. The material(s) of the shared source/drain contacts **146** can then be formed on the metal-semiconductor alloy regions **144**.

(87) In FIGS. **20**A-**20**C, a third ILD **154** is deposited over the gate spacers **102**, the second ILD **124**, the gate electrodes **134**, and the shared source/drain contacts **146**. In some embodiments, the third ILD **154** is a flowable film formed by a flowable CVD method. In some embodiments, the third ILD **154** is formed of a dielectric material such as PSG, BSG, BPSG, USG, or the like, which may be deposited by any suitable method, such as CVD, PECVD, or the like.

(88) In some embodiments, an etch stop layer (ESL) **152** is formed between the third ILD **154** and the gate spacers **102**, the second ILD **124**, the gate electrodes **134**, and the shared source/drain contacts **146**. The ESL **152** may include a dielectric material having a high etching selectivity from the etching of the third ILD **154**, such as, silicon nitride, silicon oxide, silicon oxynitride, or the like.

(89) In FIGS. **21**A-**21**C, gate contacts **156** and source/drain vias **158** are formed to contact, respectively, the gate electrodes **134** and the shared source/drain contacts **146**. The gate contacts **156** may be physically and electrically coupled to the gate electrodes **134**. The source/drain vias **158** may be physically and electrically coupled to the shared source/drain contacts **146**.

(90) As an example to form the gate contacts **156** and the source/drain vias **158**, openings for the gate contacts **156** and the source/drain vias **158** are formed through the third ILD **154** and the ESL **152**. The openings may be formed using acceptable photolithography and etching techniques. A liner (not separately illustrated), such as a diffusion barrier layer, an adhesion layer, or the like, and a conductive material are formed in the openings. The liner may include titanium, titanium nitride, tantalum, tantalum nitride, or the like. The conductive material may be cobalt, tungsten, copper, a copper alloy, silver, gold, aluminum, nickel, or the like. A planarization process, such as a CMP, may be performed to remove excess material from the top surface of the third ILD **154**. The remaining liner and conductive material form the gate contacts **156** and the source/drain vias **158** in the openings. The gate contacts **156** and the source/drain vias **158** may be formed in distinct processes, or may be formed in the same process. Although shown as being formed in the same cross-section, it should be appreciated that each of the gate contacts **156** and the source/drain vias **158** may be formed in different cross-sections, which may avoid shorting of the contacts.

(91) Features of the shared source/drain contacts **146** are illustrated in FIG. **21**B (in ghost) and in FIG. **21**C. The shared source/drain contacts **146** have upper portions **146**U in the second ILD **124** and have lower portions **146**L in the dielectric walls **72**. The upper portions **146**U are wider than the lower portions **146**L in the cross-section of FIG. **21**C (e.g., in a direction parallel to a lengthwise direction of the gate structures). In some embodiments, the lower portions **146**L have a width W.sub.1 in the range of 8 nm to 25 nm and the upper portions **146**U have a width W.sub.2 in

the range of 18 nm to 90 nm. The width W.sub.2 may be greater than the width W.sub.1. The source/drain vias **158** contact the upper portions **146**U, and forming the upper portions **146**U to be wider than the lower portions **146**L may increasing the processing window for forming the source/drain vias **158**. The lower portions **146**L are taller than the upper portions **146**U in the cross-section of FIG. **21**C. In some embodiments, the lower portions **146**L have a height H **1** in the range of 20 nm to 120 nm and the upper portions **146**U have a height H.sub.2 in the range of 10 nm to 40 nm. The height H.sub.2 may be less than the height H.sub.1. A lengthwise direction of the lower portions **146**L is parallel to a lengthwise direction of the dielectric walls **72**. In this embodiment where no losses of the dielectric walls **72** occurs during the etching of the source/drain recesses **104** (see FIGS. **10**A-**10**C), the dielectric walls **72** have a height H.sub.3 in the range of 50 nm to 250 nm and have a width W.sub.3 (between adjacent fins **62**) in the range of 10 nm to 50 nm.
(92) FIG. **21**D is a top-down view illustrated along cross-section D-D in FIGS. **21**A-**21**C. As previously described, the gate electrodes **134** extend above the dielectric walls **72** (see FIG. **21**B). As a result, the lower portions **146**L of the shared source/drain contacts **146** (e.g., the portions in the dielectric walls **72**) do not extend along the sidewalls of the gate electrodes **134** in the top-down view of FIG. **21**D. Indeed, as more clearly shown in FIG. **21**B, a majority of an shared source/drain contact **146** (e.g., the lower portion **146**L) extends along a dielectric wall **72** instead of along a gate electrode **134**, while a minority of the shared source/drain contact **146** (e.g., the upper portion **146**U) extends along the gate electrode **134**. Forming the dielectric walls **72** therefore reduces the amount of overlap between the shared source/drain contacts **146** and the gate electrodes **134**. Reducing the amount of overlap between the shared source/drain contacts **146** and the gate electrodes **134** can help reduce the parasitic capacitance between the shared source/drain contacts **146** and the gate electrodes **134**. The performance and efficiency of the resulting devices may thus be improved.
(93) The shared source/drain contacts **146** may be formed in desired locations based on the type of circuit that will be formed. In the illustrated example of FIG. **21**D, one shared source/drain contact **146** is shown. As subsequently described in greater detail, a CMOS inverter may be formed using one shared source/drain contact **146**.
(94) FIGS. **22-23** are views of complementary-FETs, in accordance with some embodiments. These embodiments are similar to the embodiment of FIG. **21**C, except losses of the dielectric walls **72** occurs during the etching of the source/drain recesses **104** (see FIGS. **10**A-**10**C), such that the height of the dielectric walls **72** is reduced. Specifically, the portions of the dielectric walls **72** beneath the gate structures (see FIG. **21**B) have a greater height than the portions of the dielectric walls **72** adjacent the upper epitaxial source/drain regions **118** and/or the lower epitaxial source/drain regions **108**.
(95) The portions of the fins **62** beneath the gate structures is shown in ghost in FIGS. **22-23**. In some embodiments, the etched portions of the dielectric walls **72** have a greater height than the portions of the fins **62** beneath the gate structures, as shown in FIG. **22**. In some embodiments, the etched portions of the dielectric walls **72** have a same height as the portions of the fins **62** beneath the gate structures, as shown in FIG. **23**. In some embodiments, the etched portions of the dielectric walls **72** have a lesser height than the portions of the fins **62** beneath the gate structures (not separately illustrated).
(96) FIGS. **24**A-**24**C are views of complementary-FETs, in accordance with some embodiments. This embodiment is similar to the embodiment of FIGS. **21**A-**21**C, except the dielectric walls **72** are formed after the STI regions **76**, such that the dielectric walls **72** are formed above the STI regions **76**. Additionally, the STI regions **76** are between each of the adjacent fins **62** and nanostructures **64**, **66**. The dielectric walls **72** contact top surfaces of some of the STI regions **76**.
(97) FIGS. **25**A-**25**C are views of complementary-FETs, in accordance with some embodiments. This embodiment is similar to the embodiment of FIGS. **21**A-**21**C, except the dielectric walls **72** are multi-layered, e.g., formed of multiple layers of different dielectric materials. For example,

each dielectric wall **72** includes a first dielectric layer **72**A and a second dielectric layer **72**B on the first dielectric layer **72**A, each of which are formed of different dielectric materials.
(98) FIGS. **26**A-**26**C are views of complementary-FETs, in accordance with some embodiments. This embodiment is similar to the embodiment of FIGS. **25**A-**25**C, except some etching of the dielectric walls **72** occurs during the formation of the openings **128** (previously described for FIGS. **16**A-**16**C). As a result, the gate electrodes **134** and/or the gate dielectrics **132** extend partially into sidewalls of the dielectric walls **72**. The gate control over the nanostructures **66** may be improved as a result of forming the gate electrodes **134** partially in the sidewalls of the dielectric walls **72**. In some embodiments, the second dielectric layers **72**B act as etch stop layers during the etching of the first dielectric layers **72**A, such that the gate electrodes **134** and/or the gate dielectrics **132** extend into/through the first dielectric layers **72**A but not the second dielectric layers **72**B.
(99) FIG. **27** is a view of complementary-FETs, in accordance with some embodiments. This embodiment is similar to the embodiment of FIG. **21**B, except only one side of each dielectric wall **72** abuts the adjacent nanostructures **66**. The other side of the dielectric walls **72** is separated from the adjacent nanostructures **66** by the gate electrodes **134**. Additionally, a dielectric wall **72** and a STI region **76** are between each pair of fins **62**. FIG. **28** is a top-down view illustrated along cross-section D-D in FIG. **27**.
(100) FIGS. **29**A-**30**C are views of additional intermediate stages in the manufacturing of complementary-FETs, in accordance with some embodiments. Additional processing of the structure of FIGS. **21**A-**21**C is described. However, the additional steps may be performed on any appropriate one of the previously described structures. FIGS. **29**A and **30**A illustrate cross-sectional views along a similar cross-section as reference cross-section A-A′ in FIG. **1**. FIGS. **29**B and **30**B illustrate cross-sectional views along a similar cross-section as reference cross-section B-B′ in FIG. **1**. FIGS. **29**C and **30**C illustrate cross-sectional views along a similar cross-section as reference cross-section C-C′ in FIG. **1**.
(101) As subsequently described in greater detail, a first interconnect structure (e.g., a front-side interconnect structure) will be formed over the substrate **50**. Some or all of the substrate **50** will then be removed and replaced with a second interconnect structure (e.g., a back-side interconnect structure). Thus, a device layer **160** of active devices is formed between a front-side interconnect structure and a back-side interconnect structure. The front-side and back-side interconnect structures each include conductive features that are connected to the devices of the device layer **160**. The conductive features (e.g., interconnects) of the front-side interconnect structure will be connected to front-sides of the upper epitaxial source/drain regions **118** and the gate electrodes **134** to form functional circuits, such as logic circuits, memory circuits, image sensor circuits, or the like. The conductive features (e.g., power rails) of the back-side interconnect structure will be connected to back-sides of the lower epitaxial source/drain regions **108** to provide a reference voltage, supply voltage, or the like to the functional circuits.
(102) In FIGS. **29**A-**29**C, a front-side interconnect structure **170** is formed on the device layer **160**, e.g., over the third ILD **154**. The front-side interconnect structure **170** is referred to as a front-side interconnect structure because it is formed at a front-side of the device layer **160** (e.g., a side of the substrate **50** on which the devices are formed, see FIGS. **21**A-**21**C). The front-side interconnect structure **170** includes dielectric layers **172** and layers of conductive features **174** in the dielectric layers **172**.
(103) The dielectric layers **172** may be formed of a dielectric material. Acceptable dielectric materials include silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), or the like, which may be formed by CVD, ALD, or the like. The dielectric layers **172** may be formed of a low-k dielectric material having a k-value lower than about 3.0. The dielectric layers **172** may be formed of an extra-low-k (ELK) dielectric material having a k-value of less than about 2.5.
(104) The conductive features **174** may include conductive lines and vias. The conductive vias may

extend through respective ones of the dielectric layers **172** to provide vertical connections between layers of conductive lines. The conductive features **174** may be formed by a damascene process, such as a single damascene process, a dual damascene process, or the like. In a damascene process, a dielectric layer **172** is patterned utilizing photolithography and etching techniques to form trenches and via openings corresponding to the desired pattern of the conductive features **174**. The trenches and via openings may then be filled with a conductive material. Suitable conductive materials include copper, aluminum, tungsten, cobalt, gold, combinations thereof, or the like, which may be formed by electroplating or the like.

(105) The front-side interconnect structure **170** includes any desired number of layers of the conductive features **174**. The conductive features **174** are connected to features of the underlying devices (e.g., the gate electrodes **134** and the upper epitaxial source/drain regions **118**) to form functional circuits. In other words, the conductive features **174** interconnect the devices of the device layer **160**.

(106) A support substrate **184** is bonded to a top surface of the front-side interconnect structure **170**. The support substrate **184** may be bonded to the front-side interconnect structure **170** by one or more bonding layer(s) **182**. The support substrate **184** may be a glass support substrate, a ceramic support substrate, a semiconductor substrate (e.g., a silicon substrate), a wafer (e.g., a silicon wafer), or the like. The support substrate **184** may provide structural support during subsequent processing steps and in the completed device. The support substrate **184** be substantially free of any active or passive devices.

(107) The support substrate **184** may be bonded to the front-side interconnect structure **170** using a suitable technique such as dielectric-to-dielectric bonding, or the like. Dielectric-to-dielectric bonding may comprise depositing the bonding layer(s) **182** on the front-side interconnect structure **170** and/or the support substrate **184**. In some embodiments, the bonding layer(s) **182** are formed of silicon oxide (e.g., a high density plasma (HDP) oxide or the like) that is deposited by CVD, ALD, or the like. The bonding layer(s) **182** may likewise include oxide layers that are formed prior to bonding using, for example, CVD, ALD, thermal oxidation, or the like. Other suitable materials may be used for the bonding layer(s) **182**.

(108) The dielectric-to-dielectric bonding process may further include applying a surface treatment to one or more of the bonding layer(s) **182**. The surface treatment may include a plasma treatment. The plasma treatment may be performed in a vacuum environment. After the plasma treatment, the surface treatment may further include a cleaning process (e.g., a rinse with deionized water, or the like) that may be applied to one or more of the bonding layer(s) **182**. The support substrate **184** is then aligned with the front-side interconnect structure **170** and the two are pressed against each other to initiate a pre-bonding of the support substrate **184** to the front-side interconnect structure **170**. The pre-bonding may be performed at about room temperature. After the pre-bonding, an annealing process may be applied. The bonds are strengthened by the annealing process. After the support substrate **184** is bonded to the front-side interconnect structure **170**, the intermediate structure is flipped so that the back-side of the device layer **160** faces upwards (not separately illustrated). The back-side of the device layer **160** refers to the side opposite to the front-side of the device layer **160**.

(109) The substrate **50** (see FIGS. **21**A-**21**C) is thinned to remove (or at least reduce the thickness of) the back-side portions of the substrate **50**. The thinning process may include a mechanical grinding, a chemical mechanical polish (CMP), an etch back, combinations thereof, or the like. The thinning process may also thin the fins **62** and the STI regions **76**. In the illustrated embodiment, the thinning process removes an entirety of the substrate **50**, the fins **62**, and the STI regions **76**, thereby exposing the gate electrodes **134**, the gate dielectrics **132**, the lower epitaxial source/drain regions **108**, and the dielectric walls **72**.

(110) In FIGS. **30**A-**30**C, a fourth ILD **190** is deposited over the back-side of the device layer **160**, such as on the exposed surfaces of the gate electrodes **134**, the gate dielectrics **132**, the lower

epitaxial source/drain regions **108**, and the dielectric walls **72**. In some embodiments, the fourth ILD **190** is a flowable film formed by a flowable CVD method. In some embodiments, the fourth ILD **190** is formed of a dielectric material such as PSG, BSG, BPSG, USG, or the like, which may be deposited by any suitable method, such as CVD, PECVD, or the like.

(111) In some embodiments, an etch stop layer (ESL) (not separately illustrated) is formed between the fourth ILD **190** and the gate electrodes **134**, the gate dielectrics **132**, the lower epitaxial source/drain regions **108**, and the dielectric walls **72**. The ESL may include a dielectric material having a high etching selectivity from the etching of the fourth ILD **190**, such as, silicon nitride, silicon oxide, silicon oxynitride, or the like.

(112) Lower source/drain contacts **194** are formed through the fourth ILD **190**. The lower source/drain contacts **194** may be physically and electrically coupled to the lower epitaxial source/drain regions **108**. Specifically, the lower source/drain contacts **194** are coupled to the back-sides of the lower epitaxial source/drain regions **108**. As an example to form the lower source/drain contacts **194**, contact openings may be formed through the fourth ILD **190** to expose the lower epitaxial source/drain regions **108**. The contact openings may be formed using acceptable photolithography and etching techniques. A liner, such as a diffusion barrier layer, an adhesion layer, or the like, and a conductive material are then formed in the contact openings. The liner may include titanium, titanium nitride, tantalum, tantalum nitride, or the like. The liner may be deposited by a conformal deposition process, such as physical vapor deposition (PVD), chemical vapor deposition (CVD), or the like. In some embodiments, the liner may include an adhesion layer and at least a portion of the adhesion layer may be treated to form a diffusion barrier layer. The conductive material may be tungsten, cobalt, ruthenium, aluminum, nickel, copper, a copper alloy, silver, gold, or the like. The conductive material may be deposited by PVD, CVD, or the like. A planarization process, such as a CMP, may be performed to remove excess material from a surface of the fourth ILD **190**. The remaining liner and conductive material in the contact openings forms the lower source/drain contacts **194**.

(113) Optionally, metal-semiconductor alloy regions **192** are formed at the interfaces between the lower source/drain contacts **194** and the lower epitaxial source/drain regions **108**. The metal-semiconductor alloy regions **192** can be silicide regions formed of a metal silicide (e.g., titanium silicide, cobalt silicide, nickel silicide, etc.), germanide regions formed of a metal germanide (e.g. titanium germanide, cobalt germanide, nickel germanide, etc.), silicon-germanide regions formed of both a metal silicide and a metal germanide, or the like. The metal-semiconductor alloy regions **192** can be formed before the lower source/drain contacts **194** by depositing a metal in the openings for the lower source/drain contacts **194** and then performing a thermal anneal process. The metal can be any metal capable of reacting with the semiconductor materials (e.g., silicon, silicon-germanium, germanium, etc.) of the lower epitaxial source/drain regions **108** to form a low-resistance metal-semiconductor alloy, such as nickel, cobalt, titanium, tantalum, platinum, tungsten, other noble metals, other refractory metals, rare earth metals or their alloys. The metal can be deposited by a deposition process such as ALD, CVD, PVD, or the like. After the thermal anneal process, a cleaning process, such as a wet clean, may be performed to remove any residual metal from the openings for the lower source/drain contacts **194**, such as from surfaces of the metal-semiconductor alloy regions **192**. The material(s) of the lower source/drain contacts **194** can then be formed on the metal-semiconductor alloy regions **192**.

(114) A back-side interconnect structure **200** is formed on the fourth ILD **190**. The back-side interconnect structure **200** is referred to as a back-side interconnect structure because it is formed at the back-side of the device layer **160**. The back-side interconnect structure **200** includes dielectric layers **202** and layers of conductive features **204** in the dielectric layers **202**.

(115) The dielectric layers **202** may be formed of a dielectric material. Acceptable dielectric materials include silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), or the like, which may be formed by CVD, ALD, or the like. The

dielectric layers **202** may be formed of a low-k dielectric material having a k-value lower than about 3.0. The dielectric layers **202** may be formed of an extra-low-k (ELK) dielectric material having a k-value of less than about 2.5.

(116) The conductive features **204** may include conductive lines and vias. The conductive vias may extend through respective ones of the dielectric layers **202** to provide vertical connections between layers of conductive lines. The conductive features **204** may be formed by a damascene process, such as a single damascene process, a dual damascene process, or the like. In a damascene process, a dielectric layer **202** is patterned utilizing photolithography and etching techniques to form trenches and via openings corresponding to the desired pattern of the conductive features **204**. The trenches and via openings may then be filled with a conductive material. Suitable conductive materials include copper, aluminum, tungsten, cobalt, gold, combinations thereof, or the like, which may be formed by electroplating or the like.

(117) The back-side interconnect structure **200** includes any desired number of layers of the conductive features **204**. The conductive features **204** form a power distribution network for the devices of the device layer **160**. Some or all of the conductive features **204** are power rails **204**P, which are conductive lines that electrically connect the shared source/drain contacts **146** and/or the lower epitaxial source/drain regions **108** to a reference voltage, supply voltage, or the like. By placing the power rails **204**P at a back-side of the device layer **160** rather than at a front-side of the device layer **160**, advantages may be achieved. For example, a gate density of the devices of the device layer **160** may be increased. Further, the back-side of the device layer **160** may accommodate wider power rails, reducing resistance and increasing efficiency of power delivery to the devices of the device layer **160**. For example, a width of the conductive features **204** may be at least twice a width of a first level conductive line (e.g., conductive line **174**L) of the front-side interconnect structure **170**.

(118) In some embodiments, the lower source/drain contacts **194** are physically and electrically coupled to the shared source/drain contacts **146**. Accordingly, the combination of a lower source/drain contact **194** and a shared source/drain contact **146** extends through the device layer **160** to couple a conductive feature **204** of the back-side interconnect structure **200** to a conductive feature **174** of the front-side interconnect structure **170**.

(119) FIGS. **31**A-**31**C are views of complementary-FETs, in accordance with some embodiments. This embodiment is similar to the embodiment of FIGS. **30**A-**30**C, except the devices are interconnected to form a CMOS circuit, e.g., an inverter. Because a complementary-FET includes a lower nanostructure-FET of a first device type and an upper nanostructure-FET of a second device type, the source/drain regions of the complementary-FET may be interconnected to form CMOS circuit having a small footprint.

(120) In this embodiment, the lower nanostructure-FET includes a lower source region **108**S and a lower drain region **108**D, and the upper nanostructure-FET includes an upper source region **118**S and an upper drain region **118**D. A first portion C **1** of FIG. **31**C is shown along cross-section C1-C1′ in FIG. **31**A, illustrating the lower source region **108**S and the upper source region **118**S. A second portion C2 of FIG. **31**C is shown along cross-section C2-C2′ in FIG. **31**A, illustrating the lower drain region **108**D and the upper drain region **118**D.

(121) A same gate structure (including a gate dielectric **132** and a gate electrode **134**) extends along three sides (e.g., a top surface, a sidewall, and a bottom surface) of the lower nanostructures **66**L of the lower nanostructure-FET and along three sides (e.g., a top surface, a sidewall, and a bottom surface) of the upper nanostructures **66**U of the upper nanostructure-FET. Accordingly, the gates of the lower and upper nanostructure-FETs are coupled together. A gate contact **156** is contact with the gate structure. A shared source/drain contact **146** is in contact with the lower drain region **108**D and the upper drain region **118**D, and is also coupled to a source/drain via **158**, e.g., an output connection. Accordingly, the drain regions of the lower and upper nanostructure-FETs are coupled together. An upper source/drain contact **196** is in contact with the upper source region **118**S, and is

coupled to one of a power node and a ground node. The upper source/drain contact **196** may be formed in a similar manner as the shared source/drain contact **146**, except the upper source/drain contact **196** is coupled to the upper source region **118**S but not the lower source region **108**S. A lower source/drain contact **194** is in contact with the lower source region **108**S, and is coupled to the other of the power node and the ground node. The resulting CMOS circuit (e.g., inverter) is shown in FIG. **31**D.

(122) Other types of circuit may be formed using complementary-FETs. The device type of the upper and lower nanostructure-FETs may depend on the desired type of circuit. In various embodiments: the lower epitaxial source/drain regions **108** are p-type while the upper epitaxial source/drain regions **118** are n-type, the lower epitaxial source/drain regions **108** are n-type while the upper epitaxial source/drain regions **118** are p-type, the lower epitaxial source/drain regions **108** and the upper epitaxial source/drain regions **118** are both n-type, or the lower epitaxial source/drain regions **108** and the upper epitaxial source/drain regions **118** are both p-type.

(123) FIGS. **32-34** are views of intermediate stages in the manufacturing of complementary-FETs, in accordance with some embodiments. FIGS. **32** and **33** are three-dimensional views showing a similar three-dimensional view as FIG. **1**. FIG. **34** illustrates a cross-sectional view along a similar cross-section as reference cross-section B-B′ in FIG. **1**. This embodiment is similar to the embodiment of FIGS. **2-21**C, except the dielectric walls **72** are formed to an increased height so that they may also be utilized as gate isolation structures.

(124) In FIG. **32**, a substrate **50** is provided. The substrate **50** may be similar to that described for FIG. **2**. A multi-layer stack **52** is formed over the substrate **50**. The multi-layer stack **52** may be similar to that described for FIG. **2**, except the multi-layer stack **52** includes an upper semiconductor layer **54**C. The upper semiconductor layer **54**C and the middle semiconductor layer **54**B may be thicker than others of the first semiconductor layers **54**A.

(125) In FIG. **33**, appropriate steps as previously described are performed to form the fins **62**, the nanostructures **64**, **66**, the dielectric walls **72**, and the STI regions **76**. The nanostructures **64** include isolation structures **64**B and isolation structures **64**C, which are thicker than others of the nanostructures **64**A. The top surfaces of the dielectric walls **72** and the isolation structures **64**C may be substantially coplanar (within process variations). The dielectric walls **72** are formed adjacent the isolation structures **64**C, and thus have a greater height than in previously described embodiments, such that the top surfaces of the dielectric walls **72** are above the top surfaces of the nanostructures **66**.

(126) In FIG. **34**, appropriate steps as previously described are performed to complete formation of the complementary-FETs. The isolation structures **64**C are removed during formation of the gate structures. The dielectric walls **72** of increased height may cut a gate structure (including a gate dielectric **132** and a gate electrode **134**) into multiple gate segments. Additionally, the top surfaces of the dielectric walls **72** and the gate structures may be substantially coplanar (within process variations). The dielectric walls **72** are disposed between adjacent gate structures. Further processing may also be performed. For example, the process described for FIGS. **29**A-**30**C may be performed to form front-side and back-side interconnect structures.

(127) Embodiments may achieve advantages. As previously noted, the shared source/drain contacts **146** have portions in the dielectric walls **72**, and the gate electrodes **134** are above the dielectric walls **72**. Because the gate electrodes **134** are above the dielectric walls **72**, a majority of an shared source/drain contact **146** extends along a dielectric wall **72** instead of along a gate electrode **134**. The amount of overlap between the shared source/drain contacts **146** and the gate electrodes **134** may thus be reduced. Reducing the amount of overlap between the shared source/drain contacts **146** and the gate electrodes **134** can help reduce the parasitic capacitance between the shared source/drain contacts **146** and the gate electrodes **134**. The performance and efficiency of the resulting devices may thus be improved.

(128) In an embodiment, a device includes: a dielectric wall; nanostructures abutting the dielectric

wall; a lower source/drain region adjoining a lower subset of the nanostructures; an upper source/drain region adjoining an upper subset of the nanostructures, the upper source/drain region oppositely doped from the lower source/drain region; and a shared source/drain contact contacting the upper source/drain region and the lower source/drain region, the shared source/drain contact extending into the dielectric wall. In some embodiments of the device, the lower source/drain region is a p-type source/drain region and the upper source/drain region is an n-type source/drain region. In some embodiments of the device, the lower source/drain region is an n-type source/drain region and the upper source/drain region is a p-type source/drain region. In some embodiments, the device further includes: an inter-layer dielectric on the upper source/drain region, the shared source/drain contact extending through the inter-layer dielectric, the shared source/drain contact having an upper portion in the inter-layer dielectric and having a lower portion in the dielectric wall, the upper portion wider than the lower portion. In some embodiments, the device further includes: an inter-layer dielectric on the upper source/drain region, the shared source/drain contact extending through the inter-layer dielectric, the shared source/drain contact having an upper portion in the inter-layer dielectric and having a lower portion in the dielectric wall, the lower portion taller than the upper portion. In some embodiments, the device further includes: an isolation region on a substrate, the nanostructures protruding from the isolation region, the dielectric wall contacting a top surface of the isolation region. In some embodiments, the device further includes: an isolation region on a substrate, the nanostructures protruding from the isolation region, the dielectric wall contacting a top surface of the substrate.

(129) In an embodiment, a device includes: a dielectric wall; first nanostructures contacting a sidewall of the dielectric wall; second nanostructures above the first nanostructures and contacting the sidewall of the dielectric wall, the second nanostructures oppositely doped from the first nanostructures; and a gate structure extending along top surfaces and sidewalls of the second nanostructures, along top surfaces and sidewalls of the first nano structures, and along a top surface and a sidewall of the dielectric wall. In some embodiments, the device further includes: a lower source/drain region adjoining the first nanostructures, the dielectric wall having a first portion adjacent the lower source/drain region, the dielectric wall having a second portion beneath the gate structure, the second portion of the dielectric wall having a greater height than the first portion of the dielectric wall. In some embodiments, the device further includes: a lower source/drain region adjoining the first nanostructures, the dielectric wall having a first portion adjacent the lower source/drain region, the dielectric wall having a second portion beneath the gate structure, the second portion of the dielectric wall having a same height as the first portion of the dielectric wall. In some embodiments of the device, a top surface of the gate structure is above a top surface of the dielectric wall. In some embodiments of the device, a top surface of the gate structure is coplanar with a top surface of the dielectric wall. In some embodiments of the device, the dielectric wall is single-layered. In some embodiments of the device, the dielectric wall is multi-layered. In some embodiments, the device further includes: a lower source region; a lower drain region, the first nano structures disposed between the lower drain region and the lower source region; an upper source region; an upper drain region, the second nanostructures disposed between the upper drain region and the upper source region; a shared source/drain contact contacting the upper drain region and the lower drain region; a lower source/drain contact contacting a back-side of the lower source region; and an upper source/drain contact contacting a front-side of the upper source region.

(130) In an embodiment, a method includes: forming a dielectric wall abutting nanostructures; etching a source/drain recess in the nanostructures; growing a lower source/drain region in the source/drain recess; growing an upper source/drain region over the lower source/drain region in the source/drain recess; depositing an inter-layer dielectric on the upper source/drain region; patterning a contact opening extending through the inter-layer dielectric and into the dielectric wall, the contact opening exposing the upper source/drain region and the lower source/drain region; and forming a source/drain contact in the contact opening. In some embodiments of the method,

patterning the contact opening includes: performing a first etch to form an upper portion of the contact opening in the inter-layer dielectric, the upper portion of the contact opening exposing a top surface of the upper source/drain region; and performing a second etch to form a lower portion of the contact opening in the dielectric wall, the lower portion of the contact opening exposing a sidewall of the upper source/drain region and a sidewall of the lower source/drain region. In some embodiments of the method, forming the dielectric wall includes: forming a plurality of dielectric walls, the nanostructures disposed between the dielectric walls; and removing a subset of the dielectric walls. In some embodiments, the method further includes: after forming the dielectric wall, forming an isolation region, the nanostructures protruding from the isolation region. In some embodiments, the method further includes: before forming the dielectric wall, forming an isolation region, the nanostructures protruding from the isolation region.
(131) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A device comprising: a dielectric wall; nanostructures abutting the dielectric wall; a lower source/drain region adjoining a lower subset of the nanostructures; an upper source/drain region adjoining an upper subset of the nanostructures, the upper source/drain region oppositely doped from the lower source/drain region; a shared source/drain contact contacting the upper source/drain region and the lower source/drain region, the shared source/drain contact extending into the dielectric wall; and an inter-layer dielectric on the upper source/drain region, the shared source/drain contact extending through the inter-layer dielectric, the shared source/drain contact having an upper portion in the inter-layer dielectric and having a lower portion in the dielectric wall, the lower portion of the shared source/drain contact being taller than the upper portion of the shared source/drain contact.
2. The device of claim 1, wherein the lower source/drain region is a p-type source/drain region and the upper source/drain region is an n-type source/drain region.
3. The device of claim 1, wherein the lower source/drain region is an n-type source/drain region and the upper source/drain region is a p-type source/drain region.
4. The device of claim 1, wherein the upper portion of the shared source/drain contact is wider than the lower portion of the shared source/drain contact.
5. The device of claim 1 further comprising: an isolation region on a substrate, the nanostructures protruding from the isolation region, the dielectric wall contacting a top surface of the isolation region.
6. The device of claim 1 further comprising: an isolation region on a substrate, the nanostructures protruding from the isolation region, the dielectric wall contacting a top surface of the substrate.
7. A device comprising: a dielectric wall; first nanostructures contacting a sidewall of the dielectric wall; second nanostructures above the first nanostructures and contacting the sidewall of the dielectric wall, the second nanostructures oppositely doped from the first nanostructures; a gate structure extending along top surfaces and sidewalls of the second nanostructures, along top surfaces and sidewalls of the first nanostructures, and along a top surface and a sidewall of the dielectric wall; and a lower source/drain region adjoining the first nanostructures, the dielectric wall having a first portion adjacent the lower source/drain region, the dielectric wall having a

second portion beneath the gate structure, the second portion of the dielectric wall having a greater height than the first portion of the dielectric wall.
8. The device of claim 7, wherein a top surface of the gate structure is above a top surface of the dielectric wall.
9. The device of claim 7, wherein a top surface of the gate structure is coplanar with a top surface of the dielectric wall.
10. The device of claim 7, wherein the dielectric wall is single-layered.
11. The device of claim 7, wherein the dielectric wall is multi-layered.
12. The device of claim 7, wherein the lower source/drain region is a lower source region, and the device further comprises: a lower drain region, the first nanostructures disposed between the lower drain region and the lower source region; an upper source region; an upper drain region, the second nanostructures disposed between the upper drain region and the upper source region; a shared source/drain contact contacting the upper drain region and the lower drain region; a lower source/drain contact contacting a back-side of the lower source region; and an upper source/drain contact contacting a front-side of the upper source region.
13. A method comprising: forming a dielectric wall abutting nanostructures, wherein forming the dielectric wall comprises: forming a plurality of dielectric walls, the nanostructures disposed between the dielectric walls; and removing a subset of the dielectric walls; etching a source/drain recess in the nanostructures; growing a lower source/drain region in the source/drain recess; growing an upper source/drain region over the lower source/drain region in the source/drain recess; depositing an inter-layer dielectric on the upper source/drain region; patterning a contact opening extending through the inter-layer dielectric and into the dielectric wall, the contact opening exposing the upper source/drain region and the lower source/drain region; and forming a source/drain contact in the contact opening.
14. The method of claim 13, wherein patterning the contact opening comprises: performing a first etch to form an upper portion of the contact opening in the inter-layer dielectric, the upper portion of the contact opening exposing a top surface of the upper source/drain region; and performing a second etch to form a lower portion of the contact opening in the dielectric wall, the lower portion of the contact opening exposing a sidewall of the upper source/drain region and a sidewall of the lower source/drain region.
15. The method of claim 13 further comprising: after forming the dielectric wall, forming an isolation region, the nanostructures protruding from the isolation region.
16. The method of claim 13 further comprising: before forming the dielectric wall, forming an isolation region, the nanostructures protruding from the isolation region.
17. The method of claim 13, wherein the contact opening has an upper portion in the inter-layer dielectric and has a lower portion in the dielectric wall, the upper portion of the contact opening being wider than the lower portion of the contact opening.
18. The method of claim 13, wherein the contact opening has an upper portion in the inter-layer dielectric and has a lower portion in the dielectric wall, the lower portion of the contact opening being taller than the upper portion of the contact opening.
19. The method of claim 13, further comprising: forming a gate structure around the nanostructures, the dielectric wall having a first portion adjacent the lower source/drain region, the dielectric wall having a second portion beneath the gate structure, the second portion of the dielectric wall having a greater height than the first portion of the dielectric wall.
20. The method of claim 13, further comprising: forming a gate structure around the nanostructures, the dielectric wall having a first portion adjacent the lower source/drain region, the dielectric wall having a second portion beneath the gate structure, the second portion of the dielectric wall having a same height as the first portion of the dielectric wall.